



Jargon Buster

Burst mode: This mode of photography enables the camera to capture a series of photos in succession for as long as the shutter release button is depressed. Burst mode rates are usually in the range of 1.5 up to 5 frames per second.

CCD sensor: The Charge Coupled Device forms the core of a digital camera's imaging capability and is the digital equivalent of the retina. It is composed of a matrix of colour sensing elements that are packed into a sensor. It is these elements that are referred to when you hear terms like '2.3 megapixel camera'. The greater the number of pixels on the CCD, the greater is the camera's effective resolution.

Focal length: This is the effective focal length of the objective lens of the camera and it is specified as a range (say, 35 mm to 70 mm). Most digital cameras feature one of two types of focusing methods—fixed and auto focus. Most low-end cameras have fixed focus lenses with preset ranges such as landscape and portrait. This delivers reasonable quality but the better option is auto focusing where cameras use infrared beams that sense the distance of the target in front of the camera and automatically adjust the focus. You see this in action even in conventional cameras when you half-depress the shutter release button, bringing the image into sharp focus.

f-stop: The ratio of the focal length of the camera's lens to the diameter of the aperture in millimeters. 'Stop' is the generic term

used for the amount of light reaching the sensor or film. The light reaching the sensor is doubled or halved by changing one neighbouring value in f-stop, shutter speed or film speed. For example, an f-stop setting of f5.6 will allow double the amount of light to reach the sensor as compared to when it's set at f8 (the next incremental step).

ISO settings: ISO is essentially the sensitivity of conventional film to light. Doubling the ISO value will double the film's sensitivity, which means that ISO 200 is twice as sensitive to light as ISO 100. Therefore, when used in relation to digital cameras, this setting mimics a conventional camera's sensitivity to light.

Metering: A process that cameras use for determining which part of the frame to assign more weightage to when focussing. Today's cameras divide the view field into zones and figure out which areas are more prominent by evaluating nearness of the subject along with the total area occupied by the target in the frame, the brightness, contrast and colour in the scene, etc. The greater the number of zones, the more accurate the focussing will be. There are three kinds of metering systems used—(i) spot metering, where the subject at the centre of the frame is assigned maximum weightage for focus; (ii) matrix metering, where the frame is divided into areas that are compared to pre-defined 'focussing situations' stored in the camera and a suitable focusing method is applied accordingly;

(iii) centre-weighted metering (the default method), where the maximum focussing weightage is assigned to the central area of the frame and is used in situations like group photographs.

Red-eye reduction: A reflection of the camera's flash when a person's pupils are dilated results in a 'red eye' in the final picture. Cameras overcome this by firing a burst of flashes, which cause the subject's pupils to contract, and thus eliminate this effect.

Shutter speed: The time for which the shutter remains open to let the light pass through the lens to the sensor or film. Shutter speeds are generally around one-fourth of a second in most digital cameras at factory settings. If you are shooting fast moving objects, the shutter speed needs to be increased, else there will be a lot of motion blur in the target object. Many cameras offer shutter speeds as high as 1/1500 second to as low as 16 seconds.

White balance: This is simply the process of matching the camera's perception of white to what is really white. For example, a white sheet of paper would appear bluish in morning light and reddish at dusk. All digital cameras now come with automatic white balance where the camera guesses the kind of white compensation necessary depending on the colour and the intensity of the ambient light. There are also manual settings such as cloudy, sunny, indoor, twilight, etc if you want greater control over the white balancing of your camera.

is comfortable to hold and is also water and dust resistant.

In the 2+ megapixel category, the Kodak EasyShare DX3600 Zoom had the best menu navigation. Thanks to the self-explanatory menu buttons, novices can get used to it very quickly. The Olympus Camedia 2100 Ultra Zoom, on the other hand, was a bit difficult to learn and you would need to spend more time with the manual. In terms of looks, however, this camera is truly an eye-catcher, though it was larger and significantly heavier than the other cameras.

Some models like Fuji FinePix 2600 Zoom and Fuji FinePix 2800 Zoom had circular buttons that could be used directionally, making it easy to navigate through the menu. Apart from being very functional, these buttons also look really good compared to standard buttons.

Performance

No matter what features and capabilities the camera boasts, the final proof lies in the image it captures. To measure the imaging performance of the camera, we used a specially compiled test scene that brought out the capability of the camera to faithfully capture colour, brightness, detail and light effects (see 'Test Process').

Scene test (details): Here we subjectively evaluated the ability of the camera to pick up the various details in our test scene.

In the sub-2.0 megapixel range, the first area of details that we looked out for in the test image was the date (16th) on the desktop calendar. In this test, the only two cameras that could pick out this detail were the Kodak EasyShare DX3215 and the Olympus Camedia C-1 Zoom. Both produced a sharp result in this test where the date

showed up clearly. None of the other cameras were able to pick up this detail—especially poor were the Logitech ClickSmart 510 and the Benq DC300. All that could be seen with them was a pixellated patch in place of the number. The one camera that did a very good job in picking up the droplets of water on the flowers in the image was the Fuji FinePix A101. Compared to the other cameras in this category, the colour reproduction was very faithful to the original. The nearest rival to the Fuji FinePix A101 was the Kodak EasyShare DX3215, which also showed faithful reproduction of colours and a good level of detail. The Logitech ClickSmart 510 performed very poorly in this test—it picked out very low overall details.

Surprisingly, the Olympus Camedia C-1 Zoom fared rather badly here. This camera falls in the higher end of the